



Fuge photo registration series face
recognition module
Communication Interface
Document



福鸽科技微信公众号

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statement

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Revision History

Version	Date	Changed Content		
		Chapter	Editor	Content
1.0	2021/5/1 4	All	FG	Initial version
1.01	2021/8/3	3.3	FG	Revision 3.3 1) Example
1.2	2021/8/2 4	4	FG	Added structural description
1.3	2021/9/1 7	5	FG	Added firmware upgrade, photo registration and error code
1.4	2021/11/ 18	4	FG	Added module assembly instructions
1.5	2021/12/ 10		FG	Added technical parameters and hardware interface definitions, etc
1.6	2021/12/ 28	6	FG	Added SN unique serial number command word
1.7	2022/4/2 8	6	FG	Added USB image transfer setting command combination And corrected product parameter description
1.8	2022/08/ 31	5	FG	Added facial feature sending function description
1.9	2022/10/ 28	5	FG	Added facial feature upload function
2.0	2024/07/ 29	—	FG	Separated the "Software Development Manual" and added photo capture function
2.1	2024/09/ 14	—	FG	Opened upload feature command 0xFB and send command 0xF7 to support color + infrared feature sending description

2.2	2024/11/ 19	—	FG	Added 0xD7 to specify UserID and name when sending photos or features; extended 0x12, 0xB1 commands; added 0x74 command to capture full-view large images
2.3	2024/11/ 26	—	FG	Merge AI-10/50 protocol, and change the document name to "Fu Ge... .."
2.4	2024/12/ 13	(一) 2/3	FG	Improved the description of the "0x12" command in AI-10/50 low power detection mode. The commands for reading 0xFA and writing 0xFB for palmar vein feature were added
2.5	2024/12/ 26	(一) 2/3	FG	Improve the description of "nid" in the "note" instruction; Increment instruction

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一、 Software Communication Protocol Docking

(一) Communication Protocol

The face recognition module is in a subordinate position, and the master controller needs to use different instructions to enable the module to complete various functions.

1. Communication message format

The basic message format for communication between the main control and the FM22x series binocular face recognition module is shown in the following table.

Basic message format

SyncWord	MsgID	Size	Data	ParityCheck
2 bytes	1 byte	2 bytes	N bytes	1 byte

The following table provides detailed descriptions of each field.

Detailed description of the basic message format fields

Fields	length	illustrate
SyncWord	2bytes	Fixed message start synchronization word 0xEF 0xAA
MsgID	1byte	Message ID (e.g. RESET)
Size	2bytes	byte Data size, in bytes
Data	N bytes	The data corresponding to the message and the parameters corresponding to the command message. 65535>=N>=0, N=0 means that this message has no parameters.
Parity Check	1 byte	The parity check code of the protocol is calculated by

		performing bit-by-bit XOR operation on the remaining bytes after removing the SyncWord part of the entire protocol.
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2. Face lock algorithm module message reception (H>>M)

The complete protocol format received by the module is shown in the following table. The master control should send commands to the module according to this format.

H>>M Protocol Format Description

name	SyncWord	MsgID	Size		Data	ParityCheck
Number	2 bytes	1 byte	2 bytes		N bytes	1 byte
content	0xEFAA	command	H	L	data	checksum

The command and data definitions are shown in the following table.

H>>M protocol format data detailed description

command (* indicates data is carried)	code	data		illustrate
		Structure	content	
RESET	0x10	none		none
GET STATUS	0x11	none		none
VERIFY*	0x12	s_msg_verify_data	pd_rightaway (1byte)	Whether to cut off the power immediately after unlocking successfully (yes:1 no:0)
			Timeout (1byte)	Unlock timeout (unit: s)

VERIFY* (Applicable to AI-10/50 module)	0x12	s_msg_Auto verify_data	at_verify (1byte)	Start low-power automatic detection (yes:1 no:0) 1=Automatically recognize and output the result after a face or QR code is detected; (Note: Palm vein recognition is only started after a face is detected) 0=Single recognition and exit this mode, at this time Timeout is the single recognition timeout. Send 0x10 RESET command, or you can directly exit this mode.
			Timeout (1byte)	Single recognition timeout (Unit: s) When a face is detected passing by, recognition starts automatically. If the person has walked away, recognition will be exited after xS. The module continues to perform low-power face detection.
VERIFY* (Applicable to FM226T-XF series and AI-10/50 module)	0x12	s_msg_verify_data	pd_rightaway (1byte)	Whether to cut off the power immediately after unlocking successfully (yes:1 no:0)
			Timeout (1byte)	Unlock timeout (unit: s)

			Max_Recognition_times (1byte) (Optional)	The number of repeated identification times is limited, ranging from 0-255. The smaller the value, the fewer times the identification is attempted, and the faster the unknown user is returned! The specific time is obtained by the user through testing. This parameter is an extended parameter. If this parameter is not sent when sending, the module defaults to 30 identification attempts.
ENROLL	0x13	s_msg_enroll_data	admin	Is it set as an administrator (yes: 1 no: 0)
			user_name (32bytes)	Enter the user name
			s_face_dir (1byte)	The direction that the user needs to enter, the specific parameters are shown in the table below "Face direction definition"
			timeout (1 byte)	Enter timeout (unit s)
ENROLL_SINGLE	0x1D	s_msg_enroll_data	admin	Is it set as an administrator (yes: 1 no: 0)
			user_name (32bytes)	Enter the user name
			s_face_dir (1byte)	Not used
			timeout (1 byte)	Enter timeout (unit s)
DELETE USER*	0x20	s_msg_deluser_data	user_id_heb (1byte) user_id_leb (1byte)	ID of the user to be deleted
DELETE ALL	0x21	None		Delete all users
GET USER INFO*	0x22	s_msg_getuser_info_data	user_id_heb (1byte)	Delete all users
			user_id_leb (1byte)	
FACE RESET	0x23			Clear entry status

		None		
MID_GET_ALL_USERID (Applicable to FM226T-XF series and AI-10/50 module)	0x24	s_msg_getuserid_index	alluserid_index (1byte)	Gets the number and ID of all registered users (Each packet contains 100 user ids. The number of alluserid_index starts from 00.)
MID_GET_ALL_USERID2 (Applicable to AI-10/50 module)	0x25	s_msg_getuserid_index	alluserid_index (1byte)	Gets the number and ID of all registered users (Each packet contains 100 user ids. The number of alluserid_index starts from 00.)
MID_ENROLL_ITG	0x26	s_msg_enroll_itg	admin	Is it set as administrator (yes:1 no:0)
			user_name (32bytes)	Enter user name
			face_dir (1byte)	Not used
			enroll_type (1 byte)	Registration type: interactive Entry or single frame entry
			enable_duplicate (1 byte)	0
			timeout (1 byte)	Entry timeout (unit s)
			Reserved (3 byte)	Reserved bit
GET VERSION	0x30	None		Get software version
INIT ENCRYPTIO*	0x50	s_msg_init_encryption_data	seed[4](4 bytes)	Master sends a random number
			mode	
MID_SET_RELEASE_ENC_KEY	0x52	s_msg_enc_key_number_data	enc_key_number [16]	Encryption sequence
MID_SET_DEBUG_ENC_KEY	0x53	s_msg_enc_key_number_data	enc_key_number [16]	Encryption sequence
MID_GET_SN	0x93	None		Device unique serial number

READ_USB_UVC_PARAMETERS	0xB0	None		USB image transmission parameter settings
SET_USB_UVC_PARAMETERS (Applicable to FM223 series & FM888 series)	0xB1	s_msg_usb_uvc_data	USB type (1byte)	USB1.1 : 0x11 USB2.0 : 0x20
			Image Rotation 180 degrees and image mirroring (1byte)	BIT0:0-no rotation, 1-rotate 180 degrees BIT1:0-no mirroring, 1-mirror flip
			JPEG image quality percentage (1byte) JPEG image quality percentage (1byte)	10 ~ 99
SET_USB_UVC_PARAMETERS (Applicable to FM226 series and AI-10/50 modules)	0xB1	s_msg_usb_uvc_data	USB Type (1byte)	USB1.1: 0x11 USB2.0: 0x20 BIT7:0-Bulk 1-ISOC (注: 0xa0-USB2.0&ISOC 0x20-USB2.0&Bulk 0x11-USB1.1&Bulk 0x91-USB1.1&ISOC)
			UVC code rate (1byte)	Uvc bit rate, unit Mbps, example: 0x18 = 24Mbps
			Quality of MJPG(1byte) MJPG quality (1byte)	10-100, such as 0x4b = 75 means 75% quality. The closer the value is to 100, the clearer the image will be, but it will be controlled by the bit rate. Therefore, for high-quality images, you need to set a high bit rate accordingly!
			Image attributes	BIT0: 1 Enable mirroring BIT1: 1 Enable 180 degree

			(1byte)	inversion
SET_FACE__LOCATION_DISPLAY	0xB5	s_msg_facelocation_display (1byte)		00: The face frame is not displayed by default 01: Display face box Note: The coordinate position when the face frame is displayed directly to the screen, the corresponding image needs to be set to mirror flipping mode by the "0xB1" command.
MID_SCAN_QR_CODE	0x70	s_msg_scan_QR	00 (1 byte)	Read QR code information
			timeout (1 byte)	Timeout period (unit: s)
MID_SNAP&UPLOAD_IMAGE_1	0x71	s_msg_snapimage	00(1byte)	Snap a photo and upload it
MID_SNAP&UPLOAD_IMAGE_2	0x71	s_msg_snapimage	Seq(1byte)	Upload photos (240*320pixel) Seq accumulates 1 (Seq starts receiving photo data from 1), each packet transmits a fixed 1024 bytes of photo data, and if the last packet is less than 1024 bytes, the actual number of remaining bytes will be used for transmission.
MID_SNAP&UPLOAD_FACEIMAGE_1	0x72	s_msg_snapfaceimage	00(1byte)	320*320pixel. After capturing a face photo, compare it locally and upload it. After judging that a frontal face photo is qualified, automatically capture, compare it locally and upload the photo. The face photo is 320*320 pixels.
MID_SNAP&UPLOAD_FACEIMAGE_2	0x72	s_msg_snapfaceimage	Seq(1byte)	After capturing a face photo, compare it locally and upload it. After judging that a frontal face photo is qualified, automatically capture, compare it locally and

				upload the photo. The face photo is 320*320 pixels.
ENROLL_SNAPFACEIMAGE	0x73	s_msg_enroll_data	admin	Is it set as an administrator? (yes:1 no:0)
			user_name (32bytes)	Enroll user name
			s_face_dir(1byte)	Reserve
			timeout (1 byte)	Reserve
MID_SNAP&UPLOAD_IMAGE_1	0x74	s_msg_snapimage_b	Seq (1byte)	Capture a full-view large picture. When Seq=0, DPI is effective.
			DPI (1byte)	Image resolution 0 - 480x640 1 - 600x800 2 - 1200x1600 (仅AI-10/50有效)
MID_SNAP&UPLOAD_IMAGE_2	0x74	s_msg_snapimage_b	Seq(1byte)	When Seq>0, upload photos Seq accumulates 1 (Seq starts receiving photo data from 1), and each packet transmits 1024 bytes of photo data. If the last packet is less than 1024 bytes, the actual number of remaining bytes will be used for transmission.
RGB_LEVEL*	0xD4	s_msg_rgb_level_info_data	RGB_LEVEL (1byte)	Color recognition threshold, ranging from 0 to 4, default to 2 (configured as 1 in 100,000 error rate when 0, 1 in 10 million error rate when 2)
			Reserve (1byte)	reserved
MID_ENROLL_WITH_PHOTO&ID_1	0xD7	s_msg_enroll_photo&id	00(1byte)	Send photos or feature registration (first package command)
			00(1byte)	
			Photo len (4bytes)(Big Endian)	Photo or feature length

(It is recommended to use the 0xF7 command first)			BioType (1 byte) 0 is normal photo mode, 1 is encrypted photo mode, 2 is normal signature mode (2048 bytes [color] or 4096 [color + infrared]), 3 is compressed signature mode (1024 bytes)	The signature file is 4100, 2052 or 1028 bytes, of which 4096, 2048 or 1024 bytes are the signature code. The initial vector value of the 4-byte crc32 (big endian) CRC32 is 0xffffffff
			UID(1byte) 1-100 (FM226) 或UID(2byte) 1-1000 (AI-10/50)	User ID specified by the developer. UID=0 means the ID is generated by the module. If FM226 is set to 1-100, it is a user-specified ID. Other values will return failure.
			Duplicate_CkFg (1byte)	Duplicate entry check flag. If set to 1, the check will be started and the query will be made in the current database. If the match with the currently issued user ID (existing user) is successful, the overwrite will be allowed. If it is repeated with other user IDs, failure will be returned. If set to 0, no duplicate entry check will be performed.
			NameLen(1byte)	Name length, no more than 20 characters. If it is 0 or there is no such field, no name is assumed. [Optional]

			Name String	User Name Content
MID_ENROLL_WITH_P HOTO&ID_2	0xD7	s_msg_enroll_p hoto&id	Seq heb (1byte)	When Seq>0, the photo or feature data transmission Seq is accumulated by 1 (Seq starts to send photo data from 1), hoto data is photo data, and the photo data packet size MTU≤ 246
			Seq leb (1byte)	
			Phphoto data[n] (n bytes)	
MID_UPGRADE_FW	0xF6	None		Start via USB flash drive Upgrade module firmware
MID_ENROLL_WITH_P HOTO_1	0xF7	s_msg_enroll_p hoto	00(1byte)	Send photos or feature registration (First package command) The feature file is 4100 or 2052 or 1028 bytes, of which 4096 or 2048 or 1024 bytes are feature codes, and the initial vector value of the 4-byte crc32 (big endian) CRC32 is 0xffffffff
			00(1byte)	
			Phphoto len (4bytes)(Big Endian)	
			BioType (1 byte) 0 is normal photo mode, 1 is encrypted photo mode, 2 is normal signature mode (2048 bytes [color] or 4096 [color + infrared]), 3 is compressed signature mode (1024 bytes)	
			Name Len(1byte) (Optional)	

		Extended parameter area: If you do not need to set a name, you do not need to send extended parameters. When the total length of the parameters received by the module exceeds 7, it is considered that there are extended parameters!		saved as 20.
			Name String (Optional)	Name string
MID_ENROLL_WITH_P HOTO_2	0xF7	s_msg_enroll_p hoto	Seq heb (1byte)	The photo or feature data transmission Seq accumulates 1 (Seq starts sending photo data from 1), hoto data is the photo data, and the photo data packet size $MTU \leq 246$
			Seq leb (1byte)	
			Phpoto data[n] (n bytes)	
MID_READ_FEATUR E_1 (Applicable to AI-10/50 module)	0xFA	s_msg_read_fe ature_data_1	seq_id_heb (1byte)	Package number (00 00)
			seq_id_leb (1byte)	
			user_id_heb (1byte)	User ID to be read (1~1000 for face users, 1001~2000 for palm print and vein)
			user_id_leb	

			(1byte)	
			feature_type (1byte)	1: Color face template 2: Color + red face template 3: Palm print and vein template
MID_READ_FEATURE_2 (Applicable to AI-10/50 module)	0xFA	s_msg_read_feature_data_2	seq_id_hib (1byte)	Packet number (each packet is 246 bytes, calculate the total number of packets, and then send from 1 and accumulate in sequence until the last packet of data is received)
			seq_id_lsb (1byte)	
			user_id_hib (1byte)	User ID to be read (1~1000 for face users, 1001~2000 for palm print and vein)
			user_id_lsb (1byte)	
			Feature data[n] (n byte)	Receive template data (246 bytes per packet, the last packet is the remaining actual length data)

MID_WRITE_FEATU RE_1 (Applicable to AI-10/50 module)	0xFB	s_msg_write_f eature_data_1	seq_id_heb (1byte)	Package serial number (00 00)
			seq_id_leb (1byte)	
			feature_type (1byte)	1: Color face template 2: Color + red face template 3: Palm print and vein template
			Name Len(1byte) (Optional)	The length of the name string is 1-20. If it exceeds 20, it will be saved as 20.
		Extended parameter area: If you do not need to set a name, you do not need to send extended parameters. When the total length of the parameters received by the module exceeds 3, it is considered that there are extended parameters!	Name String[n] (Optional)	Name string n<20
MID_WRITE_FEATU RE_2	0xFB	s_msg_write_ feature_data_2	seq_id_heb (1byte)	Packet number (each packet is 246 bytes,

(Applicable to AI-10/50 module)			seq_id_leb (1byte)	calculate the total number of packets, and then send them starting from 1 and accumulating them in sequence until the last packet of data is sent)
			feature_type (1byte)	1: Color face template 2: Color + red face template 3: Palm print and vein template
			Feature data[n] (n byte)	Template data (246 bytes per packet, the last packet is the remaining actual length data)
MID_READ_FEATURE	0xFC	s_msg_read_feature_data	checkflag	0: No duplicate checking 1: Check for duplicates (Whether to check for duplicates when sending photos for registration and sending feature mode)
DEMO MODE*	0xFE	s_msg_demo_mode_data	enable (1byte)	enable:1 disable: 0

The definition of face direction is shown in the following table.

Definition of face direction

st_face_dir (face direction)	code	illustrate
FACE_DIRECTION_UP	0x10	Enter a face facing up

FACE_DIRECTION_DOWN	0x08	Enter a face facing down
FACE_DIRECTION_LEFT	0x04	Enter a face facing left
FACE_DIRECTION_RIGHT	0x02	Enter a face facing right
FACE_DIRECTION_MIDDLE	0x01	Enter a face facing forward
FACE_DIRECTION_UNDEFINE	0x00	Undefined, default is forward

The face entry command type definitions are shown in the following table.

Definition of face input command types

enroll_type (face entry command type)	Code	Description
FACE_ENROLL_TYPE_INTERACTIVE	0x0	Interact entra
FACE_ENROLL_TYPE_SINGLE	0x1	Single frame recording

示例: 0xEF 0xAA 0x10 0x00 0x00 0x10

H>>M Protocol Example

name	SyncWord	MsgID	Size		Data	ParityCheck
conten	0xEFAA	0x10 (MID_RESET)	0x00	0x00	None	0x10

This message is the RESET message sent by the master control to the module, and the data length is 0.

3. Face lock algorithm module message sending (M>>H)

The module mainly sends three types of messages: REPLY, NOTE, and IMAGE.

1) Sending REPLY message

The complete protocol of the REPLY message sent by the module to the main control is shown in the following table.

M>>H REPLY message protocol format description

name	SyncWord	MsgID	Size		Data			ParityCheck
Numb er of	2 bytes	1 byte	2 bytes		N bytes			1 byte
conten	0xEFAA	MID_R EPLY (0x00)	H	L	s_msg_reply_data			checksum
					mid (1byte)	result (1byte)	data[0] (n-byte)	

mid indicates the task that the module is currently processing. For example, when mid = MID_ENROLL (0x13), it means that the message is the message that the module replies after processing the enroll task. The details of mid are shown in the following table.

M>>H mid definition in REPLY message or ImageOrTemplate message

mid (* indicates data is carried)	code	data[0]		Description
		Structure	Content	
MID_RESET	0x10	None		None
MID_GETSTATUS *	0x11	s_msg_repy_getst atus_data	status (1byte)	The module status includes: IDLE(0),BUSY(1),ER ROR(2),INVA LID(3)
MID_VERIFY* (user_id is only available when result is MR_SUCCESS)	0x12	s_msg_reply_verif y_data	user_id_heb (1byte)	The ID of the authenticated user
			user_id_leb (1byte)	
			user_name (32 bytes)	Username
			admin (1byte)	Is this an administrator?
			unlockStatus (1 byte)	none
MID_ENROLL*	0x13	s_msg_reply_enrol	user_id_heb	Registered user's ID

(user_id is only available when result is MR_SUCCESS)		l_data	(1byte)	The face recording status in each direction
			user_id_leb (1byte)	
			face_direction(1 byte)	
MID_ENROLL_SINGLE * (只有当 result结果为 MR_SUCCESS 时才有user_id) (user_id is only available when result is MR_SUCCESS)	0x1D	s_msg_reply_enroll_single_data	user_id_heb (1byte)	Registered user's ID
			user_id_leb (1byte)	
			face_direction(1 byte)	01 (indicates front face entry)
MID_DELUSER	0x20	None		None
MID_DELALL	0x21	None		None
MID_GETUSERINFO*	0x22	s_msg_reply_getuserinfo_data	user_id_heb (1byte)	Registered user's ID
			user_id_leb (1byte)	
			user_name (32bytes)	Name of registered user
			admin(1byte)	Is this an administrator?
MID_FACERESET	0x23	None		None
MID_GET_ALL_USERID*	0x24	s_msg_reply_all_userid_data	user_counts (1 byte)	The current packet returns the number of users. The maximum number of users in a single packet is 100 at a time, and ends when the number of users returned is less than 100.

			users_id[MAX_USER_COUNTS*2] (≤100*2 bytes)	For all registered user ids, use two consecutive bytes to store one ID, starting with the highest eight bits
MID_GET_ALL_USERS* (Applicable to AI-10/50)	0x25	s_msg_reply_all_user_id_data2	user_counts (2 bytes)	The maximum number of users to be uploaded in a single packet is 100 users at a time. When the number of returned users is less than 100, the upload ends. Reduce user_counts by 100 for each packet uploaded.
			users_id[MAX_USER_COUNTS*2] (≤100*2 bytes)	For all registered user ids, use two consecutive bytes to store one ID, starting with the highest eight bits
MID_ENROLL_ITG*(只有当 result 结果为 MR_SUCCESS 时才有 user_id)	0x26	s_msg_reply_enroll_data	user_id_hleb (1byte) user_id_leb (1byte)	Registered user's ID
MID_GET_VERSION*	0x30	s_msg_reply_version_data		Version Information
MID_INIT_ENCRYPTION	0x50	s_msg_reply_init_encryption_data	device_id[20](20 bytes)	Device id information
MID_SET_RELEASE_ENC_KEY	0x52	None		None

MID_SET_DEBUG_ENC_KEY	0x53	None		None
MID_GET_SN	0x93	Device_SN[32] (32bytes)		Unique serial number information of the device, the first 8 bytes are valid
READ_USB_UVC_PARAMETERS (Applicable to FM223 series & FM888 series)	0xB0	s_msg_usb_uvc_data	USB type (1byte)	USB1.1 : 0x11 USB2.0 : 0x20
			Image rotation 180 degrees and image mirroring (1byte)	BIT0: 0-no rotation, 1-rotate 180 degrees BIT1: 0-no mirroring, 1-mirror flip
			JPEG image quality percentage (1byte)JPEG image quality percentage (1byte)	10 ~ 99
READ_USB_UVC_PARAMETERS (Applicable to FM226 series)	0xB0	s_msg_usb_uvc_data	USB Type (1byte)	USB1.1: 0x11 USB2.0: 0x20 BIT7:0-Bulk 1-ISOC (0xa0-USB2.0&ISOC 0x20-USB2.0&Bulk 0x11-USB1.1&Bulk 0x91-USB1.1&ISOC)
			UVC code rate(1byte) UVC code rate (1byte)	Uvc的码率, 单位 Mbps, 例: 0x18 = 24Mbps Uvc bit rate, unit Mbps, example: 0x18 = 24Mbps
			Quality of MJPG(1byte) MJPG quality (1byte)	10-100, current image quality

			Image properties(1byte) Image attributes (1byte)	BIT0: 1 Enable mirroring BIT1: 1 Enable 180 degree inversion
SET_USB_UVC_PARAMETERS	0xB1	None		USB image transmission parameter setting results
SET_FACE_LOCATION_DISPLAY	0xB5	00(1byte)		Face frame display Successful setting
MID_SNAP&UPLOAD_IMAGE_1	0x71	s_msg_reply_snapimage1	result(1byte)	0 = success, others = failure
			Phpoto len (4bytes) ((Big Endian)Photo len (4 bytes)(big endian mode)	For example, 00 00 28 a3 represents the image length, which is equal to 0x28a3 in big-endian mode.
MID_SNAP&UPLOAD_IMAGE_2	0x71	s_msg_reply_snapimage2 (Note: This response is returned in the format of MsgID Size Data in the ImageOrTemplate message protocol)	Phpoto data[n] (n bytes)	Size is the length of the photo data in this packet. Each packet transmits 1024 bytes of photo data. If the last packet is less than 1024 bytes, the actual number of remaining bytes will be used for transmission. Photo data is the photo data
MID_SNAP&UPLOAD_FACEIMAGE_1	0x72	s_msg_reply_snapfaceimage1	SnapFaceresult (1byte)	0 = success, others = failure

				Phphoto len (4bytes)(Big Endian)	For example, 00 00 28 a3 represents the image length, which is equal to 0x28a3 in big-endian mode.
				UserID (2bytes) (Big Endian)	Search for local user ID. UserID is equal to 00 00, which means there is no corresponding user in the module. If it is not 0, it means there is a corresponding user. Big endian mode id_high id_low
MID_SNAP&UPL OAD_FACEIMAG E_2	0x72	s_msg_reply_snapfaceimage2 (Note: This response is returned in the format of MsgID Size Data in the ImageOrTemplate message protocol)	Phphoto data[n] (n bytes)		Size is the length of the photo data in this packet. Each packet transmits 1024 bytes of photo data. If the last packet is less than 1024 bytes, the actual number of remaining bytes will be used for transmission. Photo data is the photo data
ENROLL_SNAPFACEIMAGE* (user_id is only available when SnapFaceresult is 0)(user_id is only	0x73	s_msg_reply_enrollsnapfaceimage_data	user_id_heb (1byte)		ID of registered user
			user_id_leb (1byte)		Note: Used with 0x72 when no local user is found, this entry command will

available when SnapFaceresult is 0)				not be checked for duplicates.
			face_direction(1 byte)	Default is 0
MID_SNAP&UPL OAD_IMAGE_1	0x74	s_msg_reply_snapimage1	result(1byte)	0 = success, others = failure
			Phphoto len (4bytes)(Big Endian) Photo len (4 bytes)(big endian mode)	For example, 00 00 28 a3 represents the image length, which is equal to 0x28a3 in big-endian mode.
MID_SNAP&UPL OAD_IMAGE_2	0x74	s_msg_reply_snapimage2 (Note: This response is returned in the format of MsgID Size Data of the ImageOrTemplate message protocol)	Phphoto data[n] (n bytes)	Size is the length of the photo data in this packet. Each packet transmits 1024 bytes of photo data. If the last packet is less than 1024 bytes, the actual number of remaining bytes will be used for transmission. Photo data is the photo data
MID_UPGRADE_FW	0xF6	s_msg_reply_upgradefw	Progress (1byte)	Upgrade progress percentage
MID_ENROLL_WITH_PHOTO	0xF7	s_msg_reply_enroll_photo	Seq heb(1 byte)	Package number
			Seq leb(1 byte)	
			UserID heb(1 byte)	Registered User ID (Only after the photo is sent and the registration is successful, there will be a corresponding
			UserID leb(1 byte)	

				UserID, otherwise it will be 0)
MID_ENROLL_WITH_PHOTO	0xF7	s_msg_reply_enroll_photo	Seq heb(1 byte)	Package number
			Seq leb(1 byte)	
			UserID heb(1 byte)	Registered User ID (Only after the photo is sent and the registration is successful, there will be a corresponding UserID, otherwise it will be 0)
			UserID leb(1 byte)	
MID_RGB_LEVEL	0xD4	s_msg_reply_rgb-level_info	Result (1byte)	0=Successful; other=failure;
MID_ENROLL_WITH_PHOTO_F (It is recommended to use the 0xF7 command first)	0xD7	s_msg_reply_enroll_photo_f	Seq heb(1 byte)	Package number
			Seq leb(1 byte)	
			UserID heb(1 byte)	Registered User ID (Only after the photo is sent and the registration is successful, there will be a corresponding UserID, otherwise it will be 0)
			UserID leb(1 byte)	
MID_READ_FEATURE_1	0xFA	s_msg_reply_read_feature_data		result(1byte)
MID_READ_FEATURE_2	0xFA	s_msg_reply_read_feature_data (注: 此应答按照ImageOrTemplate消息协议格式 MsgID Size Data 返回)		Feature len (4bytes) (大端模式)
MID_WRITE_FEATURE_1	0xFB	s_msg_reply_write_feature_data_1		Feature data[n] (n bytes)
				Seq heb(1 byte)
				Seq leb(1 byte)
				UserID heb (1 byte)

MID_WRITE_FEATURE_2	0xFB	s_msg_reply_write_feature_data_2	UserID leb (1 byte) Seq heb(1 byte) Seq leb(1 byte) UserID heb (1 byte) UserID leb (1 byte)
MID_DUPLICATE_CHECK	0xFC	无	
MID_DEMOMODE	0xFE	无	

Result indicates the execution result of the command, as shown in the following table.

M>>H Result definition in REPLY message

result	code	illustrate
MR_SUCCESS	0	Success
MR_REJECTED	1	The module rejected the command
MR_ABORTED	2	The registration/verification algorithm has been terminated
MR_FAILED4_CAMERA	4	The camera failed to open
MR_FAILED4_UNKNOWNREASON	5	Unknown error
MR_FAILED4_INVALIDPARAM	6	Invalid parameters
MR_FAILED4_NOMEMORY	7	Insufficient memory
MR_FAILED4_UNKNOWNUSER	8	No registered users
MR_FAILED4_MAXUSER	9	The maximum number of users has been entered.
MR_FAILED4_FACEENROLLED	10	The face has been entered.
MR_FAILED4_LIVENESSCHECK	12	The liveness detection failed.

MR_FAILED4_TIMEOUT	13	The entry or unlock timeout
MR_FAILED4_AUTHORIZATION	14	The encryption chip Authorization failed.
MR_FAILED4_READ_FILE	19	The file reading failed.
MR_FAILED4_WRITE_FILE	20	The file writing failed.
MR_FAILED4_NO_ENCRYPT	21	The communication protocol is not
MR_FAILED4_NO_RGBIMAGE	23	The RGB image is not ready.
MR_FAILED4_JPGPHOTO_LARGE	24	The JPG photo is too big (photo registration).
MR_FAILED4_JPGPHOTO_SMALL	25	The JPG photo is too small (photo registration).
MR_FAILED4_DETECT_QR	26	QR code is scanned and decoded during the identification process.

Example: 0xEF 0xAA 0x00 0x00 0x05 0x13 0x00 0x00 0x03 0x1F 0x0A

M>>H REPLY message protocol example

name	Sync Word	MsgID	Size		Data					Parity Check
content	0xEFAA	0x00 (MID_REPLY)	0x00	0x05	s_msg_reply_data					0x0A
					0x13 (MID_ENROLL)	0x00 (MR_SUCESS)	0x00 (user_idheb)	0x03 (user_idleb)	0x1F (face_direction)	

This message indicates that this is the REPLY message that the module replies to the main control. The data part occupies 5 bytes. The entry is successful and the user ID entered is 3.

2) NOTE message sending

The main function of NOTE message is to actively return some information to the master. Currently, NOTE message is mainly sent in three situations:

- The module sends NID_READY to the main control when booting;
- The module sends face information to the main control during the input process;
- During the recognition process, the module sends face information or QR code information to the MCU.

M>>H NOTE message protocol format description

Name	SyncWord	MsgID	Size		Data		ParityCheck
Bytes	2 bytes	1 byte	2 bytes		N bytes		1 byte
Content	0xEFAA	MID_N OTE (0x01)	H	L	s_msg_note_data		checksum
					nid (1byte)	data[0] (n-bytes)	

nid indicates the execution result of the algorithm during the enroll process, as follows:

M>>H Definition of nid in NOTE message

nid (* indicates data is carried)	code	Description
NID_READY	0	Module is ready
NID_FACE_STATE*	1	Algorithm is executed successfully and returns face information_note_data_face
NID_UNKNOWNERROR	2	Unknown error
NID_OTA_DONE*	3	None
NID_EYE_STATE	4	None
NID_AUTO_VERIFY*	10	Detect status
NID_QR_DATA_FLAGS	15	Stands for QR code

The data carried by NID_FACE_STATE mainly stores face information, as follows:

M>>H Description of the Data corresponding to NID_FACE_STATE in the NOTE message

Structure	s_note_data_face							
Content	state	left	top	Right	bottom	yaw	pitch	roll
Type	int16_t	int16_t	int16_t	int16_t	int16_t	int16_t	int16_t	int16_t
Bytes	2bytes	2bytes	2bytes	2bytes	2bytes	2bytes	2bytes	2bytes

state: indicates the current state of the face, mainly including the following situations:

M>>H Note: The state description of the Data corresponding to NID_FACE_STATE in the message

Face state	cod	illustrate
FACE_STATE_NORMAL	0	Normal face
FACE_STATE_NOFACE	1	No face detected
FACE_STATE_TOOUP	2	The face is too close to the top edge of
FACE_STATE_TOODOWN	3	The face is too close to the bottom edge
FACE_STATE_TOOLEFT	4	The face is too close to the left edge of
FACE_STATE_TOORIGHT	5	The face is too close to the right edge of
FACE_STATE_FAR	6	The face is too far away and cannot be
FACE_STATE_CLOSE	7	The face is too close and cannot be
FACE_STATE_EYEBROW_OCCLUSION	8	Eyebrows are blocked
FACE_STATE_EYE_OCCLUSION	9	Eyes are blocked.
FACE_STATE_FACE_OCCLUSION	10	Face is blocked
FACE_STATE_DIRECTION_ERROR	11	Wrong direction of face recorded.
FACE_STATE_EYE_CLOSE_STATUS_OPEN_EYE	12	Eyes open state detected in closed eye
FACE_STATE_EYE_CLOSE_STATUS	13	Eyes closed state.
FACE_STATE_EYE_CLOSE_UNKNOW_STATUS	14	Unable to determine the open or closed state in closed eye mode detection.

The specific meanings of other members in s_note_data_face are as follows:

left: the distance between the face frame and the leftmost side of the image (negative number means the face frame has exceeded the leftmost side of the image)

top: the distance between the face frame and the topmost side of the image (negative number means the face frame has exceeded the top of the image)

right: the distance between the face frame and the rightmost side of the image (negative number means the face frame has exceeded the rightmost side of the image)

bottom: the distance between the face frame and the bottommost side of the image (negative number means the face frame has exceeded the bottommost side of the image)

yaw, pitch, roll indicate the rotation direction. A negative yaw indicates a left turn, a positive yaw indicates a right turn; a negative pitch indicates a head up, a positive pitch indicates a head down; a negative roll indicates a head tilt to the right, a positive roll indicates a head tilt to the left.

Example: 0xEF 0xAA 0x01 0x00 0x01 0x00 0x00

M>>H NOTE message protocol example

Name	SyncWord	MsgID	Size		Data		ParityCheck
Content	0xEFAA	0x01(MID_NOTE)	0x00	0x01	s_msg_note_data		0x00
					0x00(NID_READY)	none	

This message indicates that this is a NOTE message sent by the module to the host controller.

The data length is 1 byte and the module is ready to receive other commands.

The "data" carried by "NID_AUTO_VERIFY" mainly indicates some states that enter the detection mode. Details are as follows:

Table 5- 17 M>>H NOTE message protocol example

名称	SyncWord	MsgID	Size		Data		ParityCheck
内容	0xEFAA	0x01(MID_NOTE)	0x00	0x02	s_msg_note_data		checksum
					0x0A(NID_AUTO_VERIFY)	00 (here are three states: ①When the module receives the detection mode; ②When the module gives the	

						recognition result; ③Identify timeout.	
						01 (When the module is in detection mode, once it detects the presence of a face, it actively sends it.)	
内容	0xEFAA	0x01(MID_NOTE)	0x00	Len (≤250)	0x0F(NID_AU TO_VERIFY)	Two-dimensional code string Note: The identifiable QR code must be no more than 250 bytes of ASCII code and does not contain Chinese.	checksum

3) ImageOrTemplate message sending

The complete protocol of the Image or Template message sent by the module to the host controller is shown in the following table.

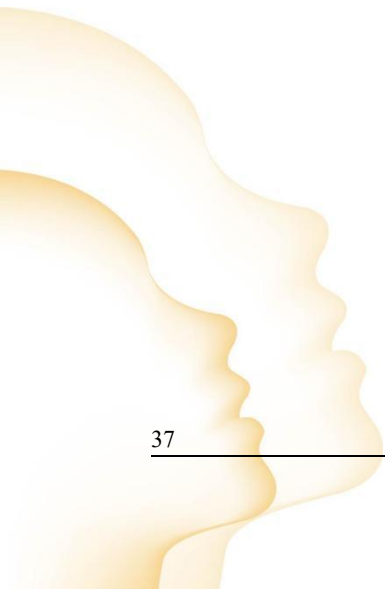
M>>H IMAGE message protocol format description

Name	SyncWord	MsgID	Size		Data	ParityCheck
Bytes	2 bytes	1 byte	2 bytes		N bytes	1 byte
Content	0xEFAA	mid	H	L	image data	checksum

mid indicates the task that the module is currently processing. For example, when mid =

MID_SNAP&UPLOAD_FACEIMAGE_2(0x72), it means that the message is the reply message after the module completes the snapshot task. For details of mid, see "Definition of mid in REPLY

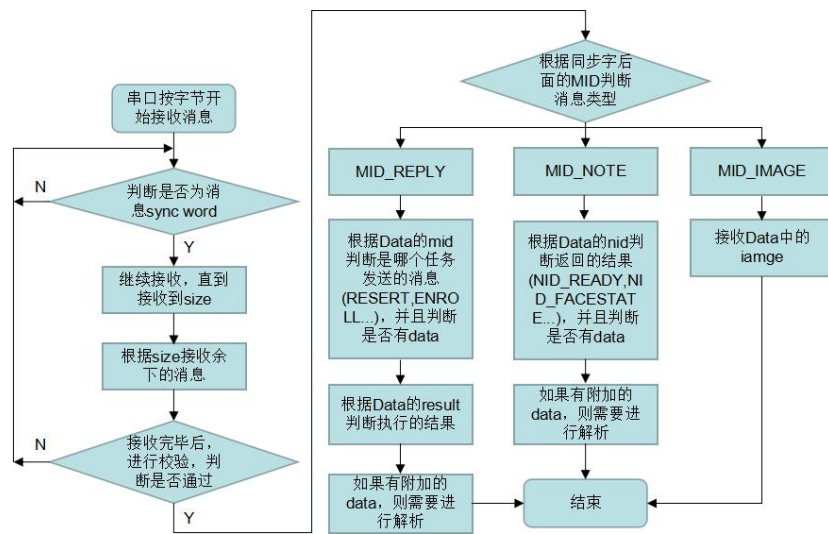
message or ImageOrTemplate message".



(二) Function Implementation Example

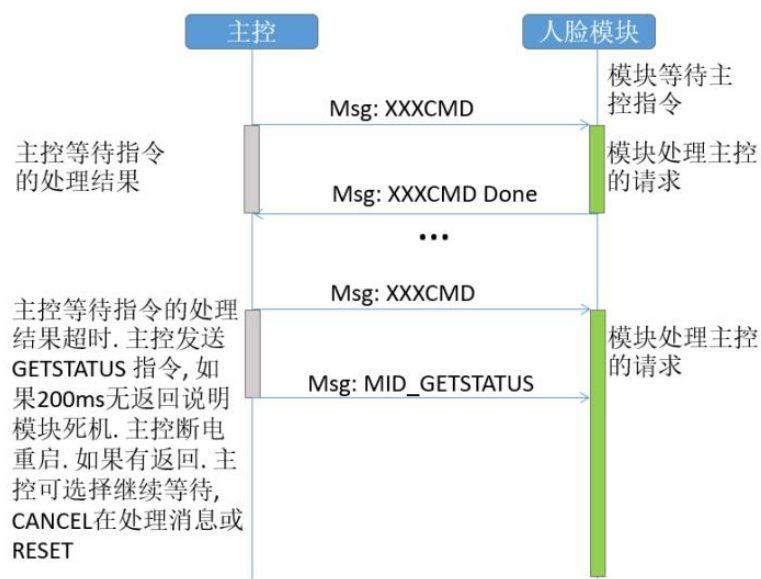
1. Master control receiving message process

The main control receives the message sent by the module according to the process shown in the figure below.



Flowchart of the main control receiving module message

2. General message processing flow



General message processing flow chart

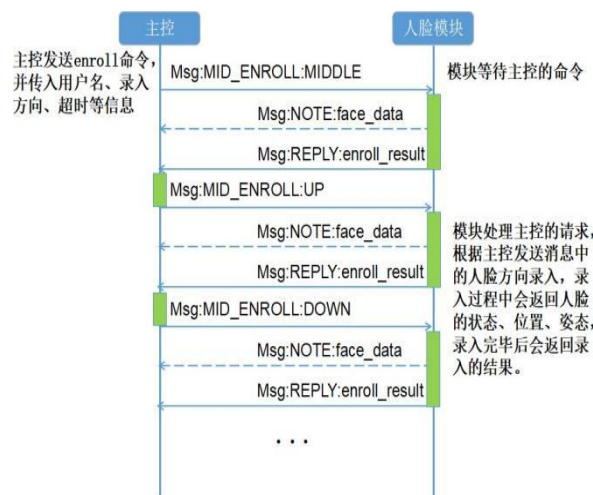
3. Power on and off process

The master control actively controls the power on and off of the module, as shown in the figure below. When the module is powered on, it enters the startup process. When it enters the standby state, it notifies the master control MSG:NOTE:READY. The master control starts to send command messages, and the module processes and returns the results. If the module has no message to send or times out, the master control can be powered off.



Power on/off processing flow chart

4. Face entry process



Face entry flow chart

During the recording process, the face lock algorithm module does not limit the order of the

recording direction. The user can arbitrarily combine the order of the recording direction. The above figure takes one of the orders as an example.

When the main control issues an instruction to record a face in a certain direction, the module starts working and returns the face status, position, posture and other information in real time (returned in the form of NOTE message). The user can use this information to remind the person recording to adjust the posture and position. When the recording is successful, the module will return the recording result in the form of REPLY message. If the recording is unsuccessful, the module will also return the reason for the failure accordingly.

It is worth noting that when the first frame image collected by the module is recorded successfully, the result will be returned directly in the form of REPLY message.

During the recording process, the FACE RESET instruction can be used to terminate the recording, and the previous recording status will also be cleared.

If the power is suddenly cut off during the recording process, the previously recorded face will not be saved.

5. Input Instructions

The angles of face recording in various directions are described as follows:

Face Entry Angle Description

Input direction	Deflection angle
Face	Facing the camera, no deflection angle
Up	Face deflected upward 5~55 degrees
Down	Face deflected downward 5~55 degrees
Right	Face deflected right 8~60 degrees
Left	Face deflected left 8~60 degrees

Among them, the upward and downward deflections are shown in the figure below.

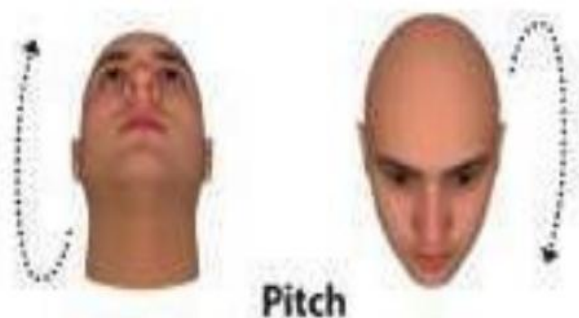
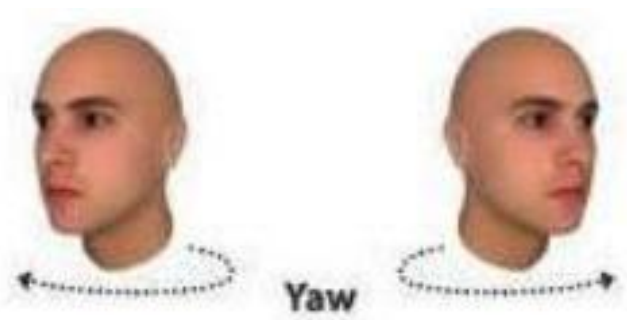


Diagram of face entry

Deflection to the left and right is shown in the figure below.



Left and right diagram of face entry

Please do not use the head rotation method shown in the following figure.



Diagram of face entry error

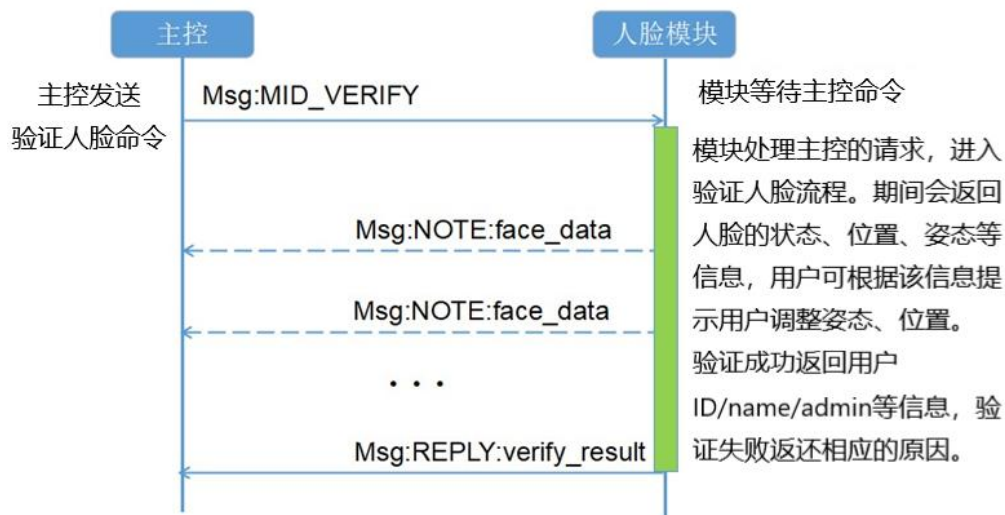
We recommend that users slowly turn their heads in a certain direction and keep the head turned during the actual data entry process. Please do not turn your head too quickly or return to the front face position immediately.

6. Single frame recording

The main difference between single-frame registration and registration process is that

single-frame registration only requires one face to be successfully registered, and does not require five interactions to complete the registration.

7. Face verification process



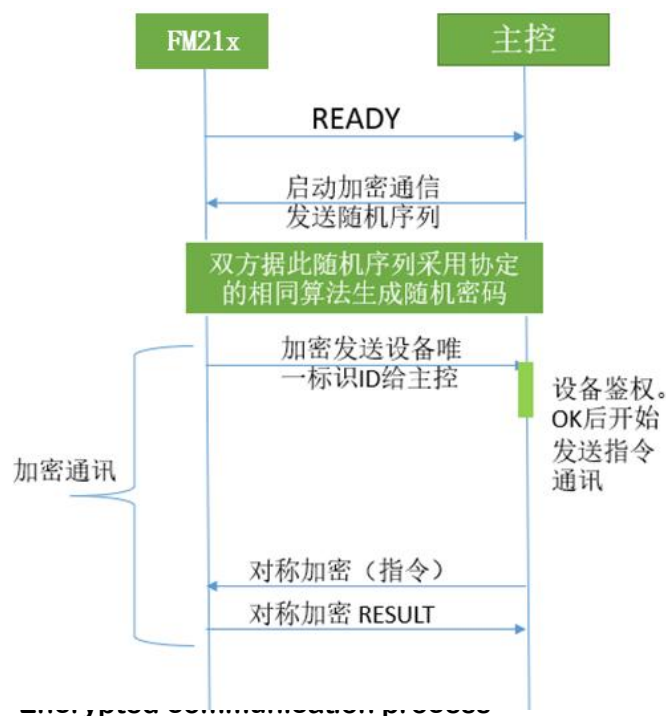
Face verification process

8. Encrypted communication process

The encrypted communication process is shown in the figure below.

- ① The master controller powers on the FM21x series face recognition module
- ② The FM21x series face recognition module sends READY (plain text) to the master controller
- ③ The first time the module is powered on, the MID_SET_RELEASE_ENC_KEY instruction needs to be called to set the 16-byte sequence required by the private protocol.
- ④ The master uses random numbers to generate a random sequence and sends it to the FM21x series face recognition module (4 bytes)
- ⑤ The FM21x series face recognition module and the master use a custom private protocol (generated a 16-byte password based on this random sequence. Both parties use the same algorithm to generate the same password. After that, both parties use this password for encryption and decryption in this session.
- ⑥ The FM21x series face recognition module returns the status and uses the random password AES/SMPL to encrypt and send the product serial number to the master

- ⑦ The master decrypts, identifies the device ID and confirms the identity before starting to process the required instructions, input or verify, etc.
- ⑧ The master uses the randomly generated password and AES/SMPL encryption to encrypt the instructions and data and send them to the FM21x series face recognition module.
- ⑨ The face recognition module decrypts the password obtained in Step 4, judges the legality of the command, and processes it.
- ⑩ The face recognition module encrypts the command processing result and data with the password decrypted in Step 4, and sends it to the main controller.



9. Photo submission registration example

MID_ENROLL_WITH_PHOTO

The main control (PC side) initiates photo registration, Seq is set to 0, Photo data is 4 bytes (big endian mode) photo size, and photo type (normal photo 0, encrypted photo 1) is 1 byte

【Seq + photo length】

The module returns Result and Seq.

When Result is MR_SUCCESS, the master controller adds 1 to Seq (Seq starts sending photo data from 1). Photo data is photo data. The size of the photo data packet is 246 bytes. When the last packet is less than 246 bytes, the actual number of bytes is sent.

When the photo is sent, the module finally returns the registration result. When Result is MR_SUCCESS, it also carries the user ID.

Example: Send a photo with a size of 2763 small photos without the user's name (Send: PC sends to module, Recv: module sends to PC)

Send:ef aa f7 00 07 00 00 00 00 0a cb 01 30

《Seq=0,photolen=0x00000acb,type=1》

Recv:ef aa 00 00 06 f7 00 00 00 00 00 f1 《result=0,Seq=0,userid=0》

Send:ef aa f7 00 f8 00 01 5b b5 01 35 7b 8e 76 99 ed 2b fe d4 7a 9e 3c de a4 6c fe d5 84 45 3c 9c a4 65 f8 d3 7c 98 39 d7 a3 6a f9 dc 72 96 36 d3 b0 60 f2 de 70 92 25 cd b7 62 ea c8 61 81 22 c2 be 71 e2 f5 5f b0 1b ff 86 41 dd c9 67 b6 0b f6 88 5d cf e1 4f aa 23 f8 9d 50 c6 e7 47 b0 0f eb 96 92 25 d5 38 9f 35 d6 ad 61 f5 d9 63 93 31 c7 96 4c e2 f4 49 ac 0e ed 96 5f cc e7 49 ac 0e ed 96 5f cc e7 49 ac 0e ed 96 5f cc e7 49 ac 0e ed 96 5f cc e7 49 ac 0e ed 96 5f cc e7 49 ac 0e ed 96 5f cc e7 49 ac c3 1f a4 7c f6 d5 03 9e 44 dc a5 4f fe d7 6a 9f 3f ce a5 92 3a d5 64 9e 3c de a1 6c ff d4 7a 9f 3d df a4 6d fe d5 7b 9e 3c de a6 6e fa d0 7d 99 34 d6 ae 66 01 11 7b 2b 2c df a6 6c fd d6 79 9a 3f da a1 69 fa d5 7b 9f 41 de a6 6e fe d1 6a 9b 2e fe 95 2c f8 c6 2a ff 92

Recv:ef aa 00 00 06 f7 00 00 01 00 00 f0

Send:ef aa f7 00 f8 00 02 3b fd d5 79 cc 54 ea 3f 34 fc e6 dc 3f c0 29 4f cc fb 97 0f 8c 57 72 94 2a c8 bc 74 e4 f0 5d b9 14 f6 8e 59 cb e3 4c a6 05 e5 e7 29 bb 93 3c d6 75 95 f7 39 ab 83 2c c6 65 85 c7 09 9b b3 1c f6 55 b5 d7 19 8b a3 0c e6 45 a5 27 e9 7b 53 fc 16 b5 55 36 fe 6a 40 ed 09 a4 46 3e cf 5d 71 de 38 9b 77 0d c7 4c 66 cf 2b 8a 68 1c d4 44 17 b8 5a f9 19 63 a5 37 1f a9 4d e8 0a 72 ba 26 0c a1 7f de 3c 40 88 18 32 93 77 d6 2e 56 9e 0a 20 8d 69 c4 26 5e 92 3a d5 64 9f 3c dc a5 6c ff d4 7a 9f 3d de a5 6d fe d5 7b 9e 3c de a6 6e fa d0 7d 99 34 d6 ae 66 01 11 7b 2b 2d df a6 6c fc d1 7f 9d 38 d8 a1 69 fa d5 7a 9c 4b df a5 6f fd c4 7f 9b 1d ee a2 7f bf 84 7c ff 4d cc 86 5f 7f dd 6f dc ad 7e 15 ac f7 f6 48 cc cc ca c6 1f 2f df 6d ba 08 3e 81 9c e9 cd 94

Recv:ef aa 00 00 06 f7 00 00 02 00 00 f3

Send:ef aa f7 00 f8 00 03 62 84 1a f8 8c 44 d4 e0 4d a9 04 e6 9e 2e ba 90 3d d9 74 96 ee 3e aa 80 2d c9 64 86 fe 0e 9a b0 1d f9 54 b6 ce 1e 8a a0 0d e9 44 a6 de ef 7d 51 fe 18 bb 57 2d e7 6c 46 ef 0b aa 48 3c f4 64 77 d8 3a 99 79 03 c5 57 7f c9 2d 88 6a 12 da 46 6c c1 5c ff 1b 61 ab 39 1d b2 54 ee 0c 70 b8 28 02 a3 47 e6 3d 47 89 1b 33 9c 76 d5 35 56 9e 0a 20 8d 69 c4 26 5e 92 24 d5 77 9d 3d df a6 7c fd c4 7b a1 3c 2b ff 02 0b 9e 8f 37 63 7e 0c 40 a5 e4 fa cd 03 95 50 a2 da a9 26 b4 5d 01 05 e6 5b 9e e7 cc 3c a1 31 38 14 bc 1e c6 df f3 48 6d 52 6e f0 6d ae ce 95 e3 13 2b 6e ad 95 c7 8b 14 6b cf 29 01 fe 54 e9 bc 54 4e 84 9e 90 23 e3 9e 54 e7 a5 d0 e3 2d 74 ed 83 ed d1 2c 60 94 3a 33 b2 2e 4a 30 75 bc 8d b1 14 5b 69 e8 e8 37 46 92 fe 83 52 51 cb b4 32 36 13

Recv:ef aa 00 00 06 f7 00 00 03 00 00 f2

Send:ef aa f7 00 f8 00 04 25 4d 60 7a 84 a2 1a e1 25 6a e6 c4 06 3b 56 a9 9d cb 27 ec 2e ee 85 3f 3e d9 88 0b 38 a4 53 c0 97 b7 d2 55 7f e5 3c 07 6f 01 a4 d0 a8 2c

53 20 bb d9 f2 b2 5d 3c c4 a7 84 c4 29 01 86 b4 04 8b d5 6a 01 14 99 a1 51 64 15 99
19 10 88 8f 9e bc be c7 a1 81 ac 8a 36 15 54 e3 fc 7b 97 59 34 e4 9d f1 0a cb 92 aa
a4 ff 65 ae 0e 8c 73 23 e7 df 83 aa b9 3c 83 64 cf 26 a2 12 f8 ad a1 6e 27 86 14 a8
fe d9 5f b3 84 e5 8f 42 47 2d 22 7b 3a 8b 4e 78 2c 0f 65 b1 d0 0c 16 26 c0 23 b7 a9
92 62 8b fb 92 3a 5b a9 6c a4 13 39 46 9b 11 29 3e db 1d 64 45 4b 5a c6 18 d3 62 ae
2f c9 4b 84 65 a2 18 be fd d1 a7 89 de 3c a2 6f 58 8c 54 9d 2a 6a c3 3f c0 5f aa 1e
9b e8 a5 a3 a7 5a b3 85 cf 9e a6 0d a3 36 ea 75 f0 9d d4 6e a1 34 ce d9 8e 9b 8c 98
2c 07 8c 6b 4e

Recv:ef aa 00 00 06 f7 00 00 04 00 00 f5

Send:ef aa f7 00 f8 00 05 80 4b b9 d8 bf 28 6f a6 34 f1 7d 22 ab 6a 53 2f 9a 98
2e 8f 26 59 45 b0 42 2b b1 76 52 74 9b 0f 73 47 2c 26 29 45 e3 44 ad a8 38 a4 74 d7
bb 75 5e 90 b9 8a b7 c9 41 c5 12 bc 62 cd f1 b7 92 bb d0 a2 20 74 46 f4 7b e3 70 4a
53 a3 d8 1e 4c 34 e1 48 ea 39 78 f2 e7 7a fb b0 6f 8c 4d 11 f4 72 94 13 a2 07 1f 3c
b0 4e 3f e0 b2 46 f9 2f 14 8f ac 85 84 47 44 af c5 89 20 e6 1b 35 75 b4 37 da 8b 5e
62 c8 bd c8 d3 04 80 12 df 82 54 4e 9f a4 73 f2 a3 88 a7 6b 0c 25 8f 78 49 c6 bd e6
94 3a 29 f4 86 f5 29 ab f8 ce a6 33 8a d2 a8 ab 38 68 54 c5 da 1f fa e3 75 1a d7 70
a9 fe b2 26 09 99 d3 f3 63 3c f6 e6 af c3 0f 6f c3 7c 90 b6 6c d2 8f 78 8e e4 c9 10
89 91 47 58 98 be cf cf e8 ac fd ff a0 11 23 11 0d a2 16 04 5a f3 b3 06 dc 45 c7 da
36 16 ca 82 f9

Recv:ef aa 00 00 06 f7 00 00 05 00 00 f4

Send:ef aa f7 00 f8 00 06 ee e3 2d f1 78 ca e6 ae a1 17 7c 8a 74 7d 93 56 46 7c
07 9d 03 14 c2 4e 91 86 2a 34 ea 13 09 e7 c5 96 fa 12 fe ff 96 fc 44 92 2d bf 10 d4
44 e8 d9 58 77 88 ad 8e 1f 7d 91 a5 74 1a b9 58 56 ba ec b9 2e 21 2f 81 36 54 4a 12
22 78 52 58 ef 8b bb dc c5 86 9c cb 18 56 d4 f5 c1 08 55 c1 68 f0 f4 6a 06 e2 64 a2
46 03 8d 20 a4 10 e4 70 99 f7 92 06 d2 b6 c6 97 01 00 c9 7e 00 6c 08 bc a7 ec 91 f6
65 84 00 af ad c0 89 77 e3 06 ea 46 b1 32 fd 98 ae c6 6c be fd 3a a9 10 f2 f9 a3 44
bb 40 d4 85 08 87 33 c0 88 bf 1b fb f2 4f 97 a2 8d 6d de 2d c2 01 89 09 36 f7 61 27
3b 3e 86 bb dc 14 f5 bf ec fa 8c 79 82 43 1c 30 24 5c 2d 78 08 6e 44 79 5f 3d 89 08
7f f3 8d 94 92 a1 70 28 c8 01 d5 87 f9 de 5f 90 56 50 87 93 1c 9f 78 e2 e9 af e2 69
a1 bc 26 9c 0c

Recv:ef aa 00 00 06 f7 00 00 06 00 00 f7

Send:ef aa f7 00 f8 00 07 d5 e1 75 5d ff f0 6d 38 51 10 3d 1e bc c0 17 38 c7 26
b3 a5 6a 81 2c 1d b6 4a 53 9c 82 a1 60 be ec 24 ee f0 8a b2 cf ea a8 b8 89 70 27 36
5a 42 77 12 f6 43 45 2b b3 a6 8c c0 16 63 17 fa 9f 55 7e 7d 55 d2 28 9c 61 9f 0a f0
c6 cc 81 73 5b 14 d8 f4 4c 99 93 b3 4f 12 3e 73 67 a5 e0 ba db a9 c0 33 99 58 e6 56
28 9f 10 c9 57 f3 86 05 c9 12 59 98 11 c6 89 de c3 5a 79 29 91 46 b8 41 74 ad 46 b5
72 b1 42 03 c3 5b f4 5d ad c7 08 0a c3 4b 9e 46 42 5c 5e 3b 55 03 05 0d 4d d8 7d 3b
23 88 07 4b 10 95 43 f3 b7 ea ad 31 0d 39 19 45 cb 1f f4 bc 96 b5 ca 8f 19 c1 28 ac
54 74 97 d4 61 cb 3f 0a 56 b2 80 dd 27 90 70 0b 4f e9 d6 62 b2 b6 64 50 a6 13 65 e5
a0 32 d6 c1 ce 89 9f 45 9a f5 1e 82 0d e1 63 ee a7 b7 e8 ea a1 52 54 7f b4 6f c2 4f
55 2e 43 21 0f

Recv:ef aa 00 00 06 f7 00 00 07 00 00 f6

Send:ef aa f7 00 f8 00 08 bd 13 35 d7 fb 09 0c b0 bc 56 0b eb 30 d0 bc 4e 53 bd

92 47 b3 91 2d 50 ed 7a d6 f8 e3 6e cd e3 d5 1a e5 56 c8 08 60 a3 85 87 f1 b9 38 ab
52 ad 52 ed 8f 46 99 40 b5 cd 38 ac a1 ae 17 7a 3a bd b7 49 e7 78 ae e7 06 da e6 cd
7d 2c 8d 65 85 db 4e 0d d8 7b fc 7d 59 67 cd d7 43 82 6a 0a 10 0e 1b 1f 8a ca 89 bf
1f b2 e2 27 13 2e 92 ac 0d 7f 55 b8 58 18 96 0d fc 2d 31 43 f6 d7 c3 df cc 0b 53 4c
3b 0e 52 c7 53 c6 c0 97 45 81 b0 4e ea ac 8d e2 80 a4 0e e4 c7 bd f2 84 8d fc 9c 5e
64 c0 dd c3 7c d4 b4 77 03 7f 9a b2 a2 67 ea 25 11 ca 30 19 c2 9f b3 28 2a 6d 37 29
d0 31 5e f4 3b df 7a 9c d1 da 74 3e b0 d8 91 e3 48 c3 e4 bb 76 0a 63 58 12 54 17 17
e8 90 d6 77 f1 74 04 73 f4 6e cc da 27 48 c0 e9 46 25 a4 f6 a6 f5 16 01 94 be d0 92
5b 49 77 7a 83

Recv:ef aa 00 00 06 f7 00 00 08 00 00 f9

Send:ef aa f7 00 f8 00 09 7e e6 67 48 27 03 9d d1 ec 51 0d 86 2c c6 8d 0f d8 de
b2 44 7d 55 48 8e 59 bd d5 f3 94 08 d7 0c 09 07 44 2c 16 7a db e4 9a 31 60 69 2d c3
e5 dc d4 91 15 18 e5 59 8d c8 d2 0a b7 f5 7b 64 e9 f7 41 6b 0f cf 18 16 e8 3b 70 9f
1a d7 4a d0 ba 06 92 7d 1e c5 b8 0e 57 fc df f4 0d 2d 90 d5 c2 0e 92 44 c0 f3 7c 86
14 9f 83 87 b8 ff ee 40 84 50 1f 1e 8e ec 5f 04 67 00 ab b7 07 59 b9 ac fc 43 28 9e
74 13 06 85 55 a3 a2 08 a0 0c df cb ad e2 2b e5 8a 18 31 c9 8b 59 44 59 7d ec 60 d6
23 5e 62 60 94 8d 27 87 1c db 43 4e 74 15 f9 5f 29 1e a4 4d 30 a5 d8 8b 8f b5 98 d8
3e 3c 37 d3 1f 9e 00 8f 5e b3 94 d6 67 eb 72 51 ad 91 49 78 ac 7f 69 4e 6d 26 16 5e
22 06 09 3a 14 fe 44 d9 73 4d fd 5c 22 fd 85 21 ee 6b bf c0 84 81 6c 26 b3 fc b4 82
af 58 d9 14 88

Recv:ef aa 00 00 06 f7 00 00 09 00 00 f8

Send:ef aa f7 00 f8 00 0a 17 bc b5 db ac 60 be 47 17 21 09 6a 18 be 22 f0 59 6a
d5 41 d3 53 66 25 23 72 c3 84 8e ac 4d 5d e8 2a 0f 1f 03 09 29 e8 46 d5 9e 4c d3 a0
62 30 37 62 6d 6f 91 e4 cc fd bd 08 87 72 78 ff 40 09 e0 c4 7a cd 68 e3 2d 6d 2b 60
fe eb b7 6b 27 0d 62 a2 dc 9f 36 20 4b 4d da 56 a1 a9 55 c5 b2 a3 16 5d 5c fa b1 53
22 9a ee 5e 15 1f ea cb 23 64 36 55 5a 8c 66 63 74 8b aa 69 f2 08 e7 72 cc d7 77 aa
88 d1 29 f8 8e a5 2e cd 5b 6d 39 fb b5 cc 24 7c 2d 6a b3 0d 5f 60 49 8e 58 37 41 b2
b9 96 78 58 0d b1 c1 40 23 8c 65 84 b9 c4 fa df 40 70 7c d6 58 07 c8 69 a8 01 11 6d
7b a1 08 d7 d4 dd 68 4a 2e bb 20 f3 fd d8 3e 3e f7 04 00 f1 46 b5 72 f4 5c 08 c1 6f
eb a6 20 bb be a1 8f ba 4a bd 58 bb ba 53 ab c1 cb 45 79 d4 a7 7f fa cb f9 0a a6 37
e0 c0 83 db be

Recv:ef aa 00 00 06 f7 00 00 0a 00 00 fb

Send:ef aa f7 00 f8 00 0b fe 81 4e 39 23 55 a4 7c 77 3e 92 05 19 08 fd 8b a6 b4
5f 28 cf a6 61 ba 31 36 12 b8 c0 0c c7 fd d0 2d 57 90 2c ae 9f cd 47 88 c5 57 5e 92
fe 20 d1 f8 95 c0 25 85 d4 71 0c 08 ef eb 25 4b a8 27 b3 c9 e7 11 ab 55 e1 74 d4 d0
d6 ce 70 9f a3 3f 30 d3 e5 f7 42 e7 b7 30 f6 fe f6 c8 6c 54 d5 62 a0 a3 47 0e fd 51 a3
f0 c7 13 df 5d 08 b8 2a 98 63 0a 18 bf ae 24 f8 9d 6f cf 67 43 07 fb e8 44 cd b4 ae
a2 a4 c1 43 9b 54 b8 7f 51 28 5e bb c0 b2 17 47 c2 3c 28 45 ec a2 51 f3 53 d6 20 d5
91 61 b0 14 7d 1c 37 c3 fd c9 89 ee 7c 38 0b b2 a8 4a ca e3 9a 13 3b da f5 8d 5b 28
8d 8e 81 9e 66 23 60 d3 c0 21 4e 9f da 44 03 28 8f 85 ef 81 4e f1 73 88 04 c0 db 4d
95 fc ed 0c d3 69 87 b6 77 f1 73 fe 7a d2 14 06 03 97 0f 56 9d ec c3 a6 71 a1 f8 9b
a9 01 f9 e5

Recv:ef aa 00 00 06 f7 00 00 0b 00 00 fa

Send:ef aa f7 00 3b 00 0c 77 02 96 95 8f fb 54 aa e2 db 08 57 9e 55 73 79 51 98 e7 0a 81 05 15 ff 23 e6 05 8d d9 b6 e6 14 76 80 72 22 ae 91 d5 e1 6c aa 45 0a 6d 9a dd 43 64 6c 91 34 ee a5 3b 92 27 24

Recv:ef aa 00 00 06 f7 00 00 0c 00 01 fc 《发送完成，UserID=1》

10.Example of QR code automatic recognition in AI-10/50 series low power automatic face detection mode.

The MCU sends the AUTO_VERIFY command, and if pd_rightaway=0, it automatically recognizes the face or the palm vein or the QR code during the recognition process. If pd_rightaway=1, the low-power automatic face detection mode is enabled, then the face and two-dimensional code will be automatically detected and recognized, and the metacarpal vein will be recognized only after the face is detected.

After the QR code is recognized and decoded successfully, the following note message is sent:

Send:ef aa 12 00 02 01 0a 1b

... ..

Recv:ef aa 01 00 02 0a 01 08

Recv:ef aa 01 00 11 01 01 00 6a 00 b4 00 fc 00 76 01 00 00 00 00 00 00 45

Recv:ef aa 01 00 59 0f 50 43 67 67 4e 33 6c 39 63 48 31 2f 42 58 41 4b 63 67 6b 45 42 33 68 74 41 48 46 72 41 48 31 30 41 51 64 35 66 33 74 6a 59 58 74 6b 64 68 55 41 48 33 59 43 63 77 59 45 66 6e 56 77 62 53 68 45 4b 44 38 66 42 48 68 37 65 47 4e 6c 65 33 67 52 43 51 6b 48 43 77 55 47 1b

Recv:ef aa 00 00 02 12 1a 0a

Recv:ef aa 01 00 02 0a 00 09

(三) Supplementary explanation

1. MID_REPLY

Each command sent by the master to the module will eventually receive a MID_REPLY reply from the module. The message contains the command mid sent by the master, the command

execution result result, and possible returned data data.

```
typedef struct {  
    uint8_t mid; // the command(message) id to reply (the request message ID)  
    uint8_t result; // command handling result: success or failed -> s_msg_result  
    uint8_t data[0];  
} s_msg_reply_data;
```

The command execution result result may be the following:

```
/* message result code */  
typedef uint8_t s_msg_result;  
const uint8_t MR_SUCCESS = 0; // success  
const uint8_t MR_REJECTED = 1; // module rejected this command  
const uint8_t MR_ABORTED = 2; // algo aborted  
const uint8_t MR_FAILED4_CAMERA = 4; // camera open failed  
const uint8_t MR_FAILED4_UNKNOWNREASON = 5; // UNKNOWN_ERROR  
const uint8_t MR_FAILED4_INVALIDPARAM = 6; // invalid param  
const uint8_t MR_FAILED4_NOMEMORY = 7; // no enough memory  
const uint8_t MR_FAILED4_UNKNOWNUSER = 8; // exceed limitation  
const uint8_t MR_FAILED4_MAXUSER = 9; // exceed maximum user number  
const uint8_t MR_FAILED4_FACEENROLLED = 10; // this face has been enrolled  
const uint8_t MR_FAILED4_LIVENESSCHECK = 12; // liveness check failed  
const uint8_t MR_FAILED4_TIMEOUT = 13; // exceed the time limit  
const uint8_t MR_FAILED4_AUTHORIZATION = 14; // authorization failed  
const uint8_t MR_FAILED4_READ_FILE = 19; // read file failed  
const uint8_t MR_FAILED4_WRITE_FILE = 20; // write file failed  
const uint8_t MR_FAILED4_NO_ENCRYPT = 21; // encrypt must be set  
const uint8_t MR_FAILED4_NO_RGBIMAGE = 23; // rgb image is not ready  
/* message result code end */
```

The returned data varies depending on the mid. In the header file message.h, the structures starting with "s_msg_reply_" are the data carried by each instruction when sending a reply message. Take verify as an example:

```
/* message reply data definitions */  
typedef struct {
```



```
uint8_t user_id_hcb;  
uint8_t user_id_lcb;  
uint8_t user_name[USER_NAME_SIZE];  
uint8_t admin;  
uint8_t unlockStatus;  
} s_msg_reply_verify_data;
```

2. MID_NOTE

MSG::NOTE is the information that the module actively reports to the main control. For example, when the module is powered on, it will actively send a NOTE to the main control, indicating that it is READY and can receive commands from the main control. After receiving the NOTE message, the main control can start sending input or verification commands.

The data contained in the NOTE message is:

```
typedef struct {  
uint8_t nid; // note id  
uint8_t data[0];  
} s_msg_note_data;  
/* module -> host note end */
```

3. MID_RESET

After the master sends this command, the module will cancel the previously executed command (such as input, unlock, etc.) and return to the STANDBY state.

4. MID_GETSTATUS

After the master sends the command, the module returns its current status, which mainly includes: MS_STANDBY (0): The module is in idle state, waiting for the master command MS_BUSY (1): The module is in working state MS_ERROR (2): The module has an error and cannot work normally MS_INVALID (3): The module is not initialized

5. MID_VERIFY

MSG::VERIFY is the most commonly used function, and the data it carries is as follows:

```
// verify
typedef struct {
    uint8_t pd_rightaway; // power down right away after verifying
    uint8_t timeout; // timeout, unit second, default 10s
} s_msg_verify_data;
```

[pd_rightaway] indicates whether to shut down the device immediately after unlocking; [timeout] is the unlocking timeout (in seconds), which is 10 seconds by default and can be set when sending the unlocking command. The timeout can be set arbitrarily by the user, with a maximum value of 255 seconds.

```
typedef struct {
    uint8_t user_id_heb;
    uint8_t user_id_leb;
    uint8_t user_name[USER_NAME_SIZE];
    uint8_t admin;
    uint8_t unlockStatus;
} s_msg_reply_verify_data;
```

6. MID_ENROLL/MID_ENROLL_SINGLE

MSG::ENROLL is also a commonly used function, and its accompanying data is as follows:

```
// enroll user
typedef struct {
    uint8_t admin; // the user will be set to admin
    uint8_t user_name[32];
    s_face_dir face_direction;
    uint8_t timeout;
} s_msg_enroll_data;
```

[admin] indicates that the person being entered is an administrator; [timeout] is the timeout during the entry process (in seconds), the default is 10 seconds, which can be set when sending the entry command. The timeout can be set by the user at will, and the maximum value does not

exceed 255 seconds; [face_direction] is only valid in MID_ENROLL, indicating the face direction that the user wants to enter. There are 5 valid face directions, defined as follows:

```
/* msg face direction */
typedef uint8_t s_face_dir;
const uint8_t FACE_DIRECTION_UP = 0x10; // face up
const uint8_t FACE_DIRECTION_DOWN = 0x08; // face down
const uint8_t FACE_DIRECTION_LEFT = 0x04; // face left
const uint8_t FACE_DIRECTION_RIGHT = 0x02; // face right
const uint8_t FACE_DIRECTION_MIDDLE = 0x01; // face middle
const uint8_t FACE_DIRECTION_UNDEFINE = 0x00; // face undefine
```

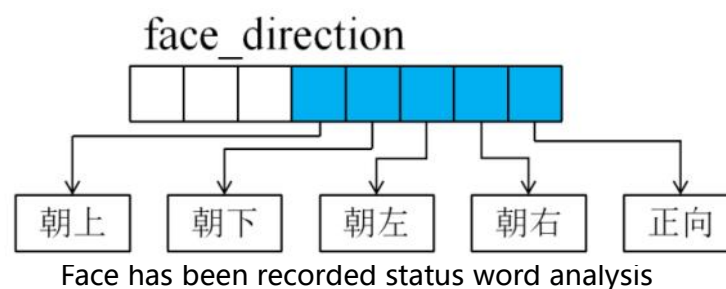
During the recording process, two messages, NOTE and REPLY, are also returned.

NOTE mainly returns the status of the face in the current frame, as well as the face position and posture.

REPLY mainly returns the final result after the recording process is completed. When MR_SUCCESS is returned, it means that the recording is successful. REPLY also returns some information, as shown below:

```
typedef struct {
    uint8_t user_id_hcb;
    uint8_t user_id_lcb;
    uint8_t face_direction;
} s_msg_reply_enroll_data;
```

Among them, [face_direction], the lower 5 bits of the data represent the face direction from high to low: up, down, left, right, and forward, as shown in the figure below. 1 means that the direction has been recorded, and 0 means that the direction has not been recorded.



7. MID_ENROLL_ITG

MSG::ENROLL_IT is a supplement and extension of the 1.15.6 ENROLL command. The entry type can be specified in the parameter structure, and the user can choose whether to repeat the entry;

The accompanying data is as follows:

```
// enroll user intergrated
```

```
typedef struct {
```

```
uint8_t admin; // the user will be set to admin, 1:admin user, 0:normal user
```

```
uint8_t user_name[USER_NAME_SIZE];
```

```
uint8_t face_direction; // reference FACE_DIRECTION_*
```

```
uint8_t enroll_type; // reference FACE_ENROLL_TYPE_*
```

```
uint8_t enable_duplicate; // enable user enroll duplicatly, 1:enable, 0:disable
```

```
    // when enroll_type is equal to FACE_ENROLL_TYPE_RGB, the enable_duplicate can be set  
to 2, it means that cant duplicate enroll with username
```

```
uint8_t timeout; // timeout unit second default 10s
```

```
uint8_t reserved[3]; // reserved feild
```

```
} s_msg_enroll_itg;
```

[enroll_type] FACE_ENROLL_TYPE_INTERACTIVE 时，为交互录入；FACE_ENROLL_TYPE_SINGLE 为单帧录入

[enable_duplicate] function is as follows:

- 0: The same person cannot be registered, but the user name can be repeated.
- 1: The same person can be registered repeatedly, and the user name can also be repeated.
- 2: The same person can be registered, but the user name cannot be repeated (only valid when registering RGB).

8. MID_DELUSER

MSG:DELUSER is used to delete a registered user, specified by user_id.

9. MID_DELALL

MSG::DELALL is to delete all registered user information.

10.MID_GETUSERINFO

The master controller specifies the registered user to be returned through the user id. After receiving the command, the module will return the information of the specified user, mainly including the following information:

```
typedef struct {  
    uint8_t user_id_heb;  
    uint8_t user_id_leb;  
    uint8_t user_name[32];  
    uint8_t admin;  
} s_msg_reply_getuserinfo_data;
```

11. MID_ENROLL_WITH_PHOTO

The master initiates photo registration, Seq is set to 0, and Photo data is 4 bytes (big endian mode) of photo size;

The module returns Result and Seq,

When Result is MR_SUCCESS, the master adds 1 to Seq (Seq starts sending photo data from 1), Photo data is photo data, and the size of photo data packets is 246 bytes. When the last packet is insufficient, the actual number of bytes is sent.

When the photo is sent, the module finally returns the registration Result, when the result is MR_SUCCESS, with the user ID.

12.MID_DEMOMODE

After the controller sends this command, the module enters the demonstration mode. In this mode, feature comparison will not be performed during the module authentication process, that is, anyone can unlock the module, but liveness detection will still be performed.

13.MID_GET_ALL_USERID

Get the number of all registered users and the IDs of all registered users.